Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

- Convert 1 eV to Joules: 1 eV = 1.6ÃĹ10^-19 Joules.

- Use formula: Î» = h / (sqrt(2 me * KE)), where h is Planck's constant (6.625ÃĹ10^-34 Js), me is electron mass (9.11ÃĹ10^-31 kg), KE = 1.6ÃĹ10^-19 J.

- Î» = 6.625ÃĹ10^-34 / sqrt(2*9.11ÃĹ10^-31*1.6ÃĹ10^-19) = 5.27 ÃĹ10^-11 m = 5.27 ÃĹ10^-9 cm, which approx to 5.27 ÃĹ10^-9 m or option F or G.

We are specifically asked for cm though, and initially 5.27ÃĹ10^-9 m = 5.27 ÃĹ10^-9ÃĹ100 cm = 5.27 ÃĹ10^-7 cm.

</think>

<answer>5.27 ÃĹ10^-7 cm</answer>